

VerdictAML

AML Alert-Triage Copilot

A defensible AI workflow for suspicious alert triage, adversarial verification, and regulator-ready STR filing.

AI assists. Analyst decides.

Verifier-backed triage

Human-in-the-loop compliance

Team

Goh Kian Xiang · Lee Zi Hao

AML teams are drowning in false positives

The analyst still has to justify every call — even though roughly 9 in 10 alerts are benign.

90-95%

AML alerts are false positives

PwC, 2019

US\$45bn

APAC financial-crime compliance
spend

LexisNexis, 2024

~5-20 min

analyst time burned per alert

Industry estimate

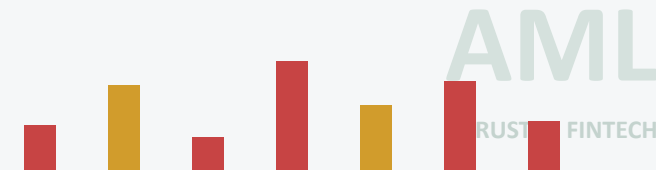
Pain

- Manual review repeats evidence gathering
- Borderline calls need a second pair of eyes
- STR drafting is slow and high-stakes

Who pays

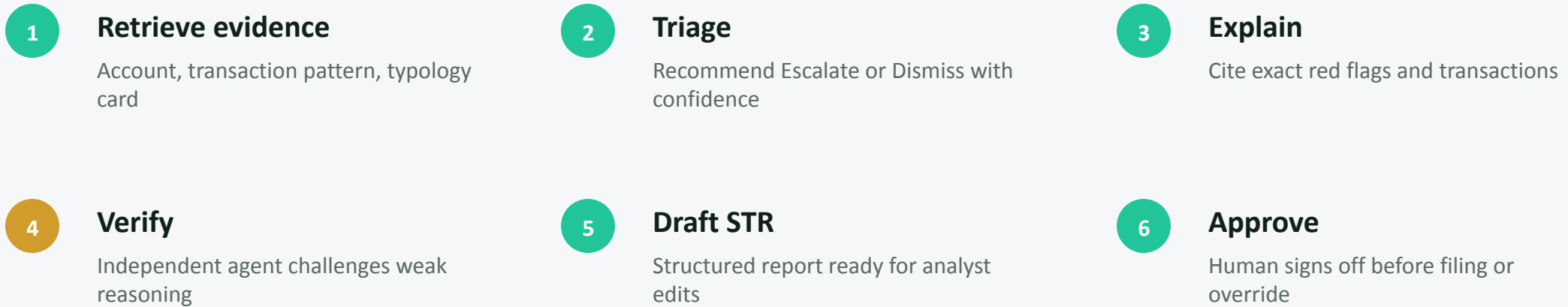
User: AML analyst
Buyer: Head of Compliance / MLRO
Risk: genuine suspicious flows get rushed

Sources: PwC (2019); LexisNexis Risk Solutions (2024).



One alert in. A defensible decision out.

The copilot handles the repetitive reasoning loop while the human remains accountable.



Positioning line

Regulator-defensible AML copilot. AI assists; analyst decides.



A second agent that challenges the first

This is what stops the copilot from being a confident-but-wrong autopilot.

HERO-001

Aisyah binti Kamal
Personal account AC-5001

Trigger

5 inbound transfers from
unrelated senders, then
forwarded onward.

Triage Agent

ESCALATE · 100% confidence

Matched FI-01 Fan-in / Fan-out.
4 of 4 indicators fired -> 100% pattern
confidence — yet the independent
verifier disagrees -> human review.

4/4 fired = full pattern match; verifier disagreement forces review

Verifier Agent

FLAGGED — disagrees

Retention of funds, partial forwarding to a
business counterparty, and consistent
inbound amounts -> legitimate merchant,
not mule consolidation.

Punch line

Triage matches the pattern. The verifier catches the benign look-alike.



Most AML tools stop at a draft. We emit the regulator's real filing.

On analyst sign-off, the approved STR exports as schema-valid goAML XML — the format Bank Negara Malaysia's FIU ingests.

Human-gated

Export unlocks only after an escalate sign-off; dismiss revokes it.

XSD-validated

Malformed reports cannot leave the system.

Config-swap, not code

Schema config points to a specific FIU registration file.

Closes the loop

Filing returns a FIU reference: MYFIU-2026-NNNNNN.

Why judges care

This turns explainable triage into a drop-in component of the existing STR submission flow.



What the analyst actually sees

The value is not the model call; it is the decision package the analyst can defend.

Evidence-backed confidence

Score is computed from fired typology indicators, not self-reported by the LLM.

Adversarial verifier

Second agent challenges weak escalation using benign look-alike and distinguishing test.

STR draft

Structured Suspicious Transaction Report narrative with cited transactions.

goAML export

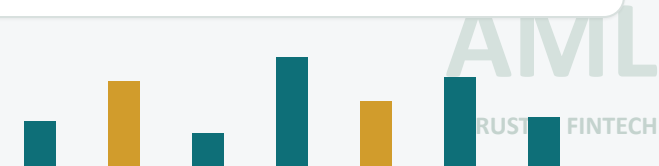
Schema-valid XML unlocks only after analyst approves an escalation.

Audit trail

Append-only events record AI recommendation, human disposition, override reason and FIU reference.

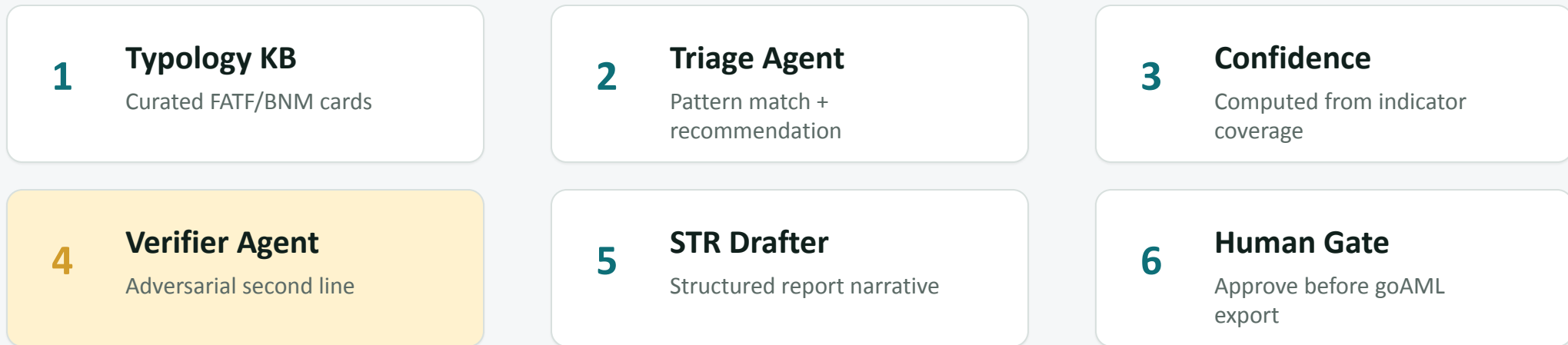
Audit-ready decisions

Recommendation, confidence, verifier status, disposition, and filing reference are recorded.



Typed agents, bounded execution

Reasoned dismissals, computed confidence, verifier challenge, and human-gated filing.

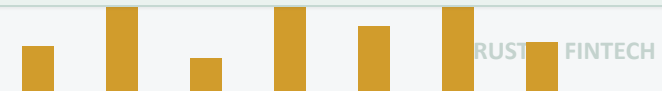


Fix to mention

No matching typology becomes a reasoned dismiss, not an endless background rerun.

Runtime resilience

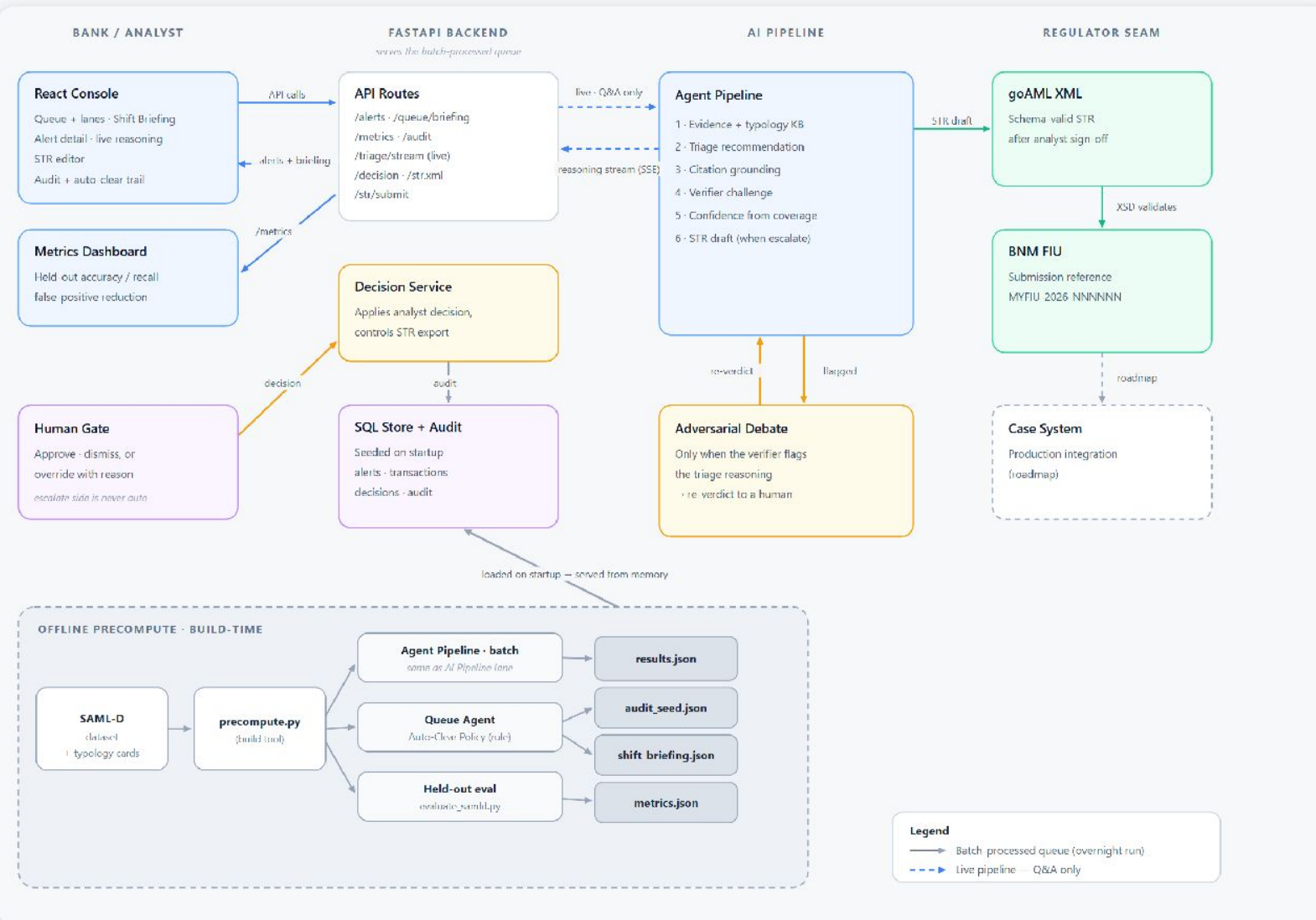
Live triage uses the same typed pipeline; provider failure returns a stable precomputed result.



ARCHITECTURE

Batch overnight, live on demand — one system end to end

An analyst console over a typed agent pipeline, durable SQL audit, and a real goAML filing seam — precomputed to scale, run live to prove it.



Batch + live

Overnight Queue Agent clears the high-confidence benign noise — and the same typed pipeline runs live on any alert, streaming its reasoning over SSE.

Durable by design

A SQLAlchemy store + append-only audit trail — alerts, transactions, decisions, audit. Real persistence, not a demo in-memory dict.

Human-gated filing

The approved STR exports as schema-valid goAML XML only after an analyst signs off — XSD-validated before it can ever leave the system. AI assists; the analyst decides.

Two public datasets — each used for what it can prove

Real amounts + counterparties drive the metric; SynthAML adds timing + scale.

SAML-D · Oztas et al., 2023

- Measurement + demo set (held-out n=250)
- Real amount + counterparty + channel
- FI-01 / ST-01 / PT-01 can finally fire
- £/€ flows = the live demo queue

SynthAML · Jensen et al., 2023

- Scale + timing reference
- 20k alerts / 16M+ transactions
- Powers DA-01 dormant-then-active
- Amount-less, no counterparty (limitation)

How we use it honestly

4 of 5 FATF/BNM cards measured; KYC-01 is the residual. HERO cases are crafted on top, separate from metrics.

Source: Oztas et al. (2023), SAML-D · Jensen et al. (2023), Nature Scientific Data / SynthAML.



Honest metrics, not inflated metrics

Measured on real SAML-D — where fan-in and structuring can finally fire. On imbalanced data, recall beats accuracy.

~68% less

Review time per alert

14 -> 4.5 min

72%

Catch-rate (recall)

@ 75% precision

68.8%

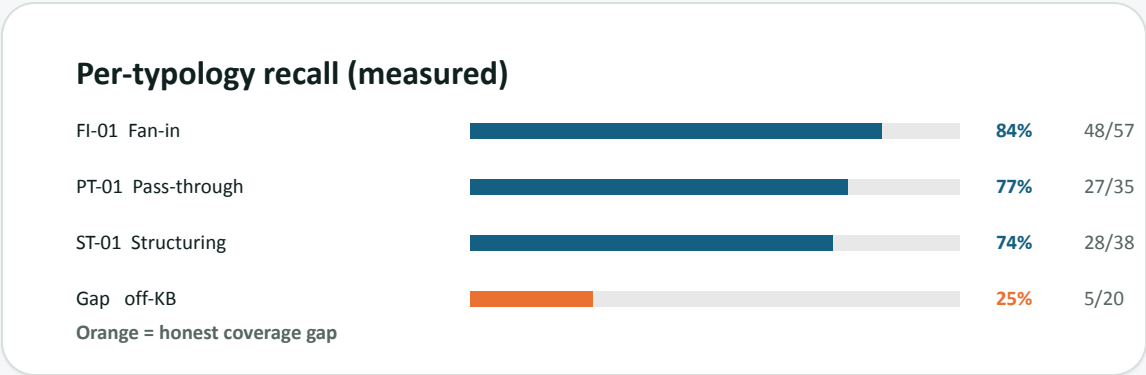
Accuracy vs labels

+28.8 pts vs baseline

60.4%

Auto-clear precision

42.4% queue auto-cleared



The honest line

Real SAML-D carries amount + counterparty, so fan-in and structuring are measurable for the first time. 4 of 5 FATF/BNM cards measured; KYC-01 is the honest residual.

Confusion (n=250): TP 108 · FP 36 · FN 42 · TN 64

Source: Measured on held-out SAML-D (Oztas et al., 2023), n=250, ADR-0012. Coverage: 4 of 5 typologies measured. Time figures modeled/illustrative.

Beachhead: compliance teams with real AML pressure

Sell time recovered, auditability, and deployment control.

Target market

Tier-2 banks, digital banks, e-wallets, remittance and fintech payment platforms in Malaysia/ASEAN.

Buyer

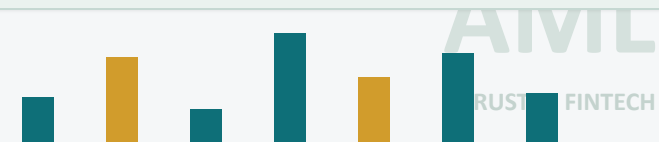
Head of Compliance / MLRO.
The user is the front-line AML analyst.

Adoption path

Shadow-mode pilot, then annual seat + consumption subscription.

Why now

Digital payment volume is rising; analysts are scarce; regulators expect explainable, auditable decisions.



Subscription tied to analyst time saved

Keep pricing simple in the pitch: pilot first, then annual Team or Enterprise contract.

Pilot / Proof of Value

8-12 week shadow-mode engagement

One-time setup fee, creditable toward annual contract

Team

<= 15 analysts, small/mid compliance teams

RM400-600 per analyst per month, billed annually

Enterprise

Banks with high alert volume and governance needs

Annual platform license + per-alert volume tier

ROI framing

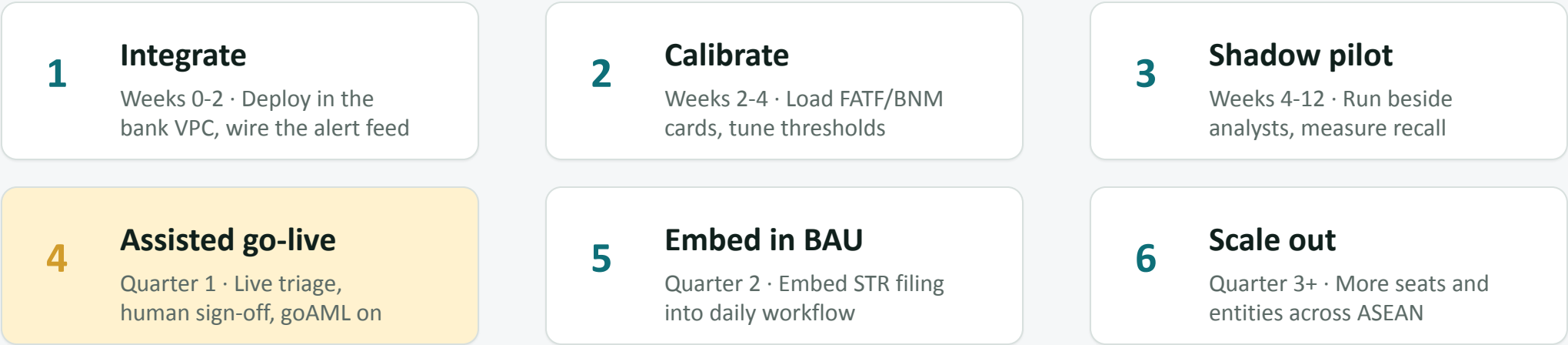
Price below a fraction of analyst cost; value comes from time returned and audit-ready consistency.

Source: Product README pricing model; indicative prelim pricing.



From pilot to embedded production

A staged rollout that earns trust before it automates, keeping adoption risk low.



Low-risk adoption

Shadow mode proves recall before a single decision is automated.

Pricing by phase

Pilot fee in shadow mode, then annual Team or Enterprise seats.



Built to run inside the bank's perimeter

The bank stays the reporting institution of record — and its data never leaves.

Data stays inside

Production runs on-prem or in the bank's private VPC.

Provider-agnostic

LLM sits behind one swappable client; self-hosted model is a config change.

No customer-data training

No labeled-data demand, retraining loop, drift monitoring, or hidden model-risk box.

Capital-light

No GPU training farm; enterprise software team can run it economically on-prem.

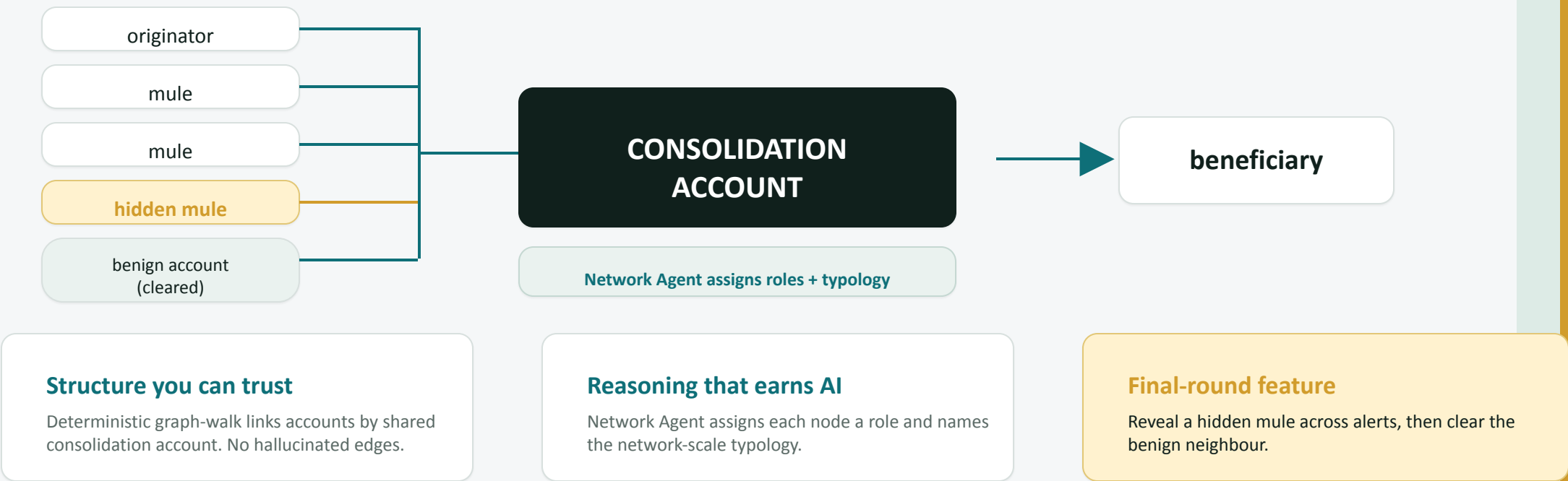
Deployment line

Hosted DeepSeek proves reasoning; production deployment can be private by design.



From one account to the whole network

Today the copilot reasons one alert at a time. Next: lift it from the account to the cluster.



Honest line Account-level triage has a recall ceiling. Link-analysis breaks it. Roadmap capability; no metric change.

► LIVE DEMO

Now watch it run

VerdictAML — the real pipeline, on real SAML-D alerts

Why VerdictAML is memorable

- Real workflow pain: AML alert overload
- Technical depth: multi-agent evidence, verifier, goAML export
- Commercial path: compliance teams can pilot in shadow mode
- Human control: AI assists; analyst remains accountable

Two-person team

Goh Kian Xiang

AI & Backend Engineer

Lee Zi Hao

Frontend Engineer

One-liner

Full-stack AI reasoning plus a polished analyst console.